PAVAN DHARMA ADAPA

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PROFESSIONAL SUMMARY

Healthcare Data Analyst with an M.S. in Computer Science & Engineering and hands-on expertise designing production-grade ETL pipelines, dimensional (star-schema) data warehouses, and clinical-operations dashboards across SQL, Python, Tableau, and Power BI. Adept at translating clinical and operational requirements into validated data models, patient registries, and provider-facing reports that mirror real-world Epic Clarity, Cogito, and Caboodle reporting workflows. Proven analyzing 900K+ healthcare records and engineering analytics across ED throughput, 30-day readmission risk, revenue cycle, and ambulatory operations. Grounded in EHR/EMR concepts, HL7, ICD-10/CPT, and HIPAA.

TECHNICAL SKILLS

Epic Ecosystem & Reporting (Self-Study / Certification Prep): Clarity data model (ENCOUNTER, PAT_ENC, ORDER_PROC, RESULT, PAT_DIAGNOSIS), Cogito, Caboodle, Reporting Workbench concepts, SlicerDicer, Radar dashboards; report development with Crystal Reports, SSRS (SQL Server Reporting Services), and SAP BusinessObjects; Epic Sphinx assessment prep.

Healthcare Data & Standards: EHR/EMR systems, clinical & claims data, revenue cycle (denials, A/R, charge capture), population health, patient registries, HL7 v2, ICD-10 / CPT coding, HIPAA compliance

Data Engineering & Governance: ETL / ELT pipeline development, data warehousing, dimensional / star-schema modeling, data validation, data cleansing & standardization, data quality, data governance

Programming & Querying: SQL (joins, CTEs, window functions, query optimization), Python (Pandas, NumPy), PostgreSQL, MySQL, Advanced Excel (Pivot Tables, VLOOKUP)

BI & Visualization: Tableau, Microsoft Power BI, Matplotlib, Seaborn — interactive dashboards, KPI reporting, executive-ready visualization

Analytics & Predictive Modeling: Hypothesis testing, exploratory data analysis (EDA), risk stratification, predictive modeling (logistic regression, Random Forest, XGBoost), model evaluation

Tools & Cloud: Git, Linux, AWS, Jupyter Notebook, Selenium

PROFESSIONAL EXPERIENCE

Graduate Teaching Assistant & Analytics Mentor Sep 2025 – May 2026

Mississippi State University — Starkville, MS

- Trained and supported **100+ students** across 6 courses on data cleaning, transformation, visualization, and dashboard-based reporting in SQL, Python (Pandas, Matplotlib), and Advanced Excel, building practical analyst-workflow competency.
- Enabled early identification of at-risk users by tracking KPIs (completion rates, performance distribution, retention) in SQL and Excel, improving intervention turnaround.
- Built and led hands-on workshops on IoT sensor data for the SmartCT smart-city initiative, producing traffic, energy, and environmental KPI dashboards while reinforcing **data-quality and validation practices**.
- Created reusable training materials and documentation that improved task-completion speed by 25% and standardized analytics best practices across cohorts.

Graduate Research Analyst — Healthcare Data Analytics Aug 2024 – Aug 2025

Mississippi State University — Starkville, MS

- Designed and automated a production ETL pipeline (Python, Selenium, SQL) to extract, cleanse, and standardize multi-source clinical and surveillance datasets, **cutting 95% data-processing time** (from several hours to under 10 minutes per dataset) and improving downstream reporting reliability.
- Partnered with cross-functional stakeholders to translate clinical and operational questions into technical data requirements, querying **900,000+ public-health surveillance records** in SQL and Python to surface 8 key drivers of disease transmission and deliver data-backed recommendations to non-technical decision-makers.
- Built reproducible, enterprise-scale reporting by engineering dimensional / star-schema data models and enforcing field-level **data validation, standardization, and governance** across multiple source systems.
- Developed risk-stratification and predictive models (logistic regression, Random Forest, XGBoost) reaching **87% forecast accuracy**, and established baseline performance KPIs to monitor model and data quality over time.
- Reduced manual analysis time 50% and resolved 35% of dataset inconsistencies by automating feature-engineering and data-cleansing routines, **strengthening data integrity** for production dashboards.
- Delivered presentation-ready Tableau and Power BI dashboards and correlation analyses that translated complex findings into actionable insight for leadership decision-making.

HEALTHCARE ANALYTICS PROJECTS

Portfolio projects are built on synthetic, non-PHI data modeled on Epic's publicly documented Clarity / Caboodle schemas and reporting patterns.

ED Throughput & Patient Flow Analytics — Clinical Operations

- Replicated **Clarity-style schema** (ENCOUNTER, ORDER_PROC, PAT_DIAGNOSIS) on a 5,000-visit synthetic ED dataset to analyze clinical workflows, quantifying a 4–6 PM peak where median door-to-bed time nearly doubled (≈20 to 39 min) and left-without-being-seen (LWBS) rose from 5.1% to 6.6% — building the staffing case for peak-hour coverage.
- Engineered a dimensional data warehouse and Python/SQL ETL pipeline with derived flow metrics (door-to-bed, bed-to-provider, length of stay) feeding interactive Tableau and Power BI **provider dashboards** for operations leadership.
- Flagged the 5 highest-readmission diagnoses for care-management follow-up by computing KPIs across 10 diagnosis groups (LWBS 5.5%, admit rate 24%, 30-day readmission 15.5%).

Multimodal LLM Evaluation (LLaVA-1.5) – Analytics & Experimentation

- Designed and implemented evaluation pipelines to analyze cross-lingual performance of multimodal large language models (LLaVA-1.5), applying NLP techniques, data contamination auditing, and statistical analysis to compare model outputs across multiple benchmarks.
- Aggregated experiment metrics in Python and SQL, building clean analytical datasets and visualization-ready summaries that enabled systematic comparison of models, hyperparameters, and prompts, and supported data-driven recommendations on model selection.
- Presented findings to academic stakeholders through data storytelling, KPI-driven summaries, and visualization-rich reports, demonstrating strong stakeholder management, cross-functional collaboration, and the ability to translate complex technical results into clear, actionable insights for non-technical audiences.

30-Day Readmission Risk Model & Patient Registry — Population Health

- Built and tuned a logistic-regression 30-day readmission model (**0.73 AUC**) on a 2,000-patient synthetic registry, capturing ~40% of actual readmissions within the top 20% of flagged patients (2x lift over baseline) to prioritize care-management outreach.
- Implemented an automated daily SQL + Python scoring pipeline (feature standardization; low/medium/high risk binning) feeding a Tableau "Daily Risk Board" that supported proactive 48-hour pre-discharge intervention.
- Structured a reusable **patient registry** and validation checks mirroring Clarity / Caboodle reporting patterns to support reproducible population-health analytics.

Revenue Cycle & Denials Analytics Dashboard — Resolute / Claims

- Analyzed 8,000+ synthetic claims modeled on **Epic Resolute** Professional & Hospital Billing data to identify the top 5 denial reason codes (CARC/RARC) driving ~60% of denied charges, prioritizing rework for the highest-recovery accounts.
- Engineered a SQL/Python ETL pipeline transforming charge, transaction, and remittance (835/EOB-style) data into a **dimensional revenue-cycle data mart** for repeatable reporting.
- Built a Tableau/Power BI revenue-cycle dashboard tracking **clean claim rate, days in A/R, denial rate, and net collection rate** by payer and service area, surfacing a payer segment with elevated denials for targeted follow-up.

Ambulatory Access & Provider Productivity Analytics — EpicCare Ambulatory

- Analyzed 12,000+ synthetic ambulatory encounters modeled on **EpicCare Ambulatory** clinical workflows to compute access and productivity KPIs — third-next-available appointment, an 18% no-show rate, and provider utilization — across 15 simulated clinics.
- Built chronic-disease **patient registries** (diabetes, hypertension) and care-gap flags supporting population-health outreach and quality-measure tracking.
- Delivered provider- and clinic-level dashboards (visit volume, panel size, wRVU productivity) in Tableau/Power BI, enabling clinic operations leaders to rebalance schedules and reduce access bottlenecks.

EDUCATION

M.S., Computer Science & Engineering — Mississippi State University, MS May 2026

Relevant Coursework: Advanced Machine Learning, Design Parallel Algorithms, Artificial Intelligence, Algorithms, Advance Cyber Operations, Software engineering, and Data analytics

B. Tech, Computer Science & Engineering — Malla Reddy University, India May 2024

Relevant Coursework: Data Structures and Algorithms, Operating Systems, Database Management Systems (DBMS), Computer Networks, and Object-Oriented Programming

PUBLICATION (IN PROGRESS)

Computational Tools for Modeling and Predicting Respiratory Disease Spread in Commercial Poultry Production

Zhang Li, Pavan Dharma Adapa, Sai Harika Gade, Saikanth Ratnavale, Navicky Michael.

- **Designed data-driven forecasting frameworks** by synthesizing compartmental (SIR/SEIR) and machine-learning models to predict respiratory disease transmission across large-scale populations.
- **Assessed the efficacy of advanced statistical methods** (Random Forest, deep learning) in identifying disease risk factors and detecting anomalies from remote sensor and surveillance data.